COS 433: Cryptography

Announcements:

- Problem Set 4 due Wed 11/11
- Check-in 2 this week

Digital signatures (authentication)

Digital Signatures



$\mathrm{Gen}(1^\lambda) \to (\mathrm{sk}, \mathrm{vk})$



m

vk (broadcast)

$(m, \sigma = \mathrm{Sign}(\mathrm{sk}, m))$



$\mathrm{Verify}(\mathrm{vk}, m, \sigma) \in \{0, 1\}$

Definition: A digital signature scheme with message length $n = n(\lambda)$ consists of efficient algorithms $\mathrm{Gen}, \mathrm{Sign}, \mathrm{Verify}$. It must satisfy the following properties:

- **Correctness:** for all m, $\mathrm{Verify}(\mathrm{vk}, m, \mathrm{Sign}(\mathrm{sk}, m)) = 1$ with probability 1.
- Existential Unforgeability: for all efficient oracle adversaries $A^{\mathrm{Sign}(\mathrm{sk},\cdot)}$, we have:

$$\Pr_{(\mathrm{sk},\mathrm{vk})}\left[A^{\mathrm{Sign}(\mathrm{sk},\cdot)}(\mathrm{vk}) \to (m^*, \sigma^*) : m^* \text{ not queried and } \mathrm{Verify}(\mathrm{vk}, m^*, \sigma^*) = 1\right] = \mathrm{negl}(\lambda)$$

Digital Signatures



$\text{Gen}(1^\lambda) \rightarrow (\text{sk}, \text{vk})$



$m \in \{0,1\}^n$

vk (broadcast)

$(m, \sigma = \text{Sign}(\text{sk}, m))$



$\text{Verify}(\text{vk}, m, \sigma) \in \{0, 1\}$

Notes:

- Message length matters. Not clear how to go from signing one-bit messages to signing longer messages.
- Oracle-based security definition is still required, since the signing key is secret.

$$\Pr_{(\text{sk},\text{vk})}\left[A^{\text{Sign}(\text{sk},\cdot)}(\text{vk}) \rightarrow (m^*, \sigma^*) : m^* \text{ not queried and } \text{Verify}(\text{vk}, m^*, \sigma^*) = 1\right] = \text{negl}(\lambda)$$

“Textbook” RSA Signatures

$\mathrm{sk} = (N, d)$
$\mathrm{vk} = (N, e)$





(N, e)

$(m, \sigma = m^d \bmod N)$



$m \in \mathbb{Z}_N^\times$

$\mathrm{Verify}((N, e), m, \sigma) = 1$ if $\sigma^e \equiv m \pmod N$

Notes:

- Restricted to $m \in \mathbb{Z}_N^\times$
- Does not satisfy existential unforgeability. Why?
  - Can multiply any two signatures together to get a new one!
- Secure if the queries and challenge m^* are all random and outside the adversary’s control.

“Textbook” RSA Signatures

$\mathrm{sk} = (N, d)$
$\mathrm{vk} = (N, e)$



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$\mathrm{Verify}((N, e), m, \sigma) = 1$ if $\sigma^e \equiv m \pmod N$

Random Message Unforgeability: for all efficient adversaries $A^{\mathrm{Sign}(\mathrm{sk},\cdot)}$ of the form



m^*, vk

“help me”

$m, \mathrm{Sign}(\mathrm{sk}, m)$



$$\Pr_{(\mathrm{sk},\mathrm{vk})}\left[A^{\mathrm{Sign}(\mathrm{sk},\perp)}(m^*, \mathrm{vk}) \to \sigma^* : \mathrm{Verify}(\mathrm{vk}, m^*, \sigma^*) = 1\right] = \mathrm{negl}(\lambda)$$

“Textbook” RSA Signatures

$\mathrm{sk} = (N, d)$
$\mathrm{vk} = (N, e)$



$m \in \mathbb{Z}_N^\times$



(N, e)

$(m, \sigma = m^d \bmod N)$



$\mathrm{Verify}((N, e), m, \sigma) = 1$ if $\sigma^e \equiv m \pmod N$

Proof of random message unforgeability: we want



m^*, N, e

”help me”

m, m^d



$$\Pr_{(\mathrm{sk},\mathrm{vk})}\left[A^{\mathrm{Sign}(\mathrm{sk},\perp)}(m^*, N, e) \to (m^*)^d\right] = \mathrm{negl}(\lambda)$$

“Textbook” RSA Signatures

$\mathrm{sk} = (N, d)$
$\mathrm{vk} = (N, e)$





$m \in \mathbb{Z}_N^\times$

(N, e)

$(m, \sigma = m^d \bmod N)$



$\mathrm{Verify}((N, e), m, \sigma) = 1$ if $\sigma^e \equiv m \pmod N$

Key Fact:

$$(m, m^d) \equiv (\sigma^e, \sigma)$$

“Textbook” RSA Signatures

$\mathrm{sk} = (N, d)$
$\mathrm{vk} = (N, e)$





(N, e)

$(m, \sigma = m^d \bmod N)$



$m \in \mathbb{Z}_N^\times$

$\mathrm{Verify}((N, e), m, \sigma) = 1$ if $\sigma^e \equiv m \pmod N$

Proof of random message unforgeability: equivalently, we want



m^*, N, e

”help me”

σ^e, σ



sk not needed here!

Breaks RSA assumption

$$\Pr_{(\mathrm{sk},\mathrm{vk})}\left[A^{\mathrm{Sign}(\mathrm{sk},\perp)}(m^*, N, e) \to (m^*)^d\right] = \mathrm{negl}(\lambda)$$

Digital Signatures



$\text{Gen}(1^\lambda) \rightarrow (\text{sk}, \text{vk})$



$m \in \{0,1\}^n$

vk

$(m, \sigma = \text{Sign}(\text{sk}, m))$



$\text{Verify}(\text{vk}, m, \sigma) \in \{0, 1\}$

Building full-fledged digital signatures is pretty difficult!

- Practical constructions are usually heuristically secure
- But we do know how to provably build signature schemes (under assumptions)

Digital Signatures



$\text{Gen}(1^\lambda) \rightarrow (\text{sk}, \text{vk})$



$m \in \{0,1\}^n$

vk

$(m, \sigma = \text{Sign}(\text{sk}, m))$



$\text{Verify}(\text{vk}, m, \sigma) \in \{0, 1\}$

Today and next two classes:

1. One-time signatures (easier to build)
2. RSA signatures (heuristic)
3. DSA/Schnorr signatures (heuristic)

One-time signatures

One-Time Signatures



$\mathrm{Gen}(1^\lambda) \to (\mathrm{sk}, \mathrm{vk})$



$m \in \{0,1\}^n$

vk (broadcast) →

$(m, \sigma = \mathrm{Sign}(\mathrm{sk}, m))$ →



$\mathrm{Verify}(\mathrm{vk}, m, \sigma) \in \{0,1\}$

Definition: A one-time digital signature scheme with message length $n = n(\lambda)$ consists of efficient algorithms $\mathrm{Gen}, \mathrm{Sign}, \mathrm{Verify}$. It must satisfy **correctness** as well as:

- One-time Existential Unforgeability: for all efficient oracle adversaries $A^{\mathrm{Sign}(\mathrm{sk},\cdot)}$ that query the signing oracle once, we have:

$$\Pr_{(\mathrm{sk},\mathrm{vk})}\left[A^{\mathrm{Sign}(\mathrm{sk},\cdot)}(\mathrm{vk}) \to (m^*, \sigma^*) : m^* \text{ not queried and } \mathrm{Verify}(\mathrm{vk}, m^*, \sigma^*) = 1\right] = \mathrm{negl}(\lambda)$$

One-Time Signatures



$\mathrm{Gen}(1^\lambda) \to (\mathrm{sk}, \mathrm{vk})$



$m \in \{0,1\}^n$

vk (broadcast)

$(m, \sigma = \mathrm{Sign}(\mathrm{sk}, m))$



$\mathrm{Verify}(\mathrm{vk}, m, \sigma) \in \{0, 1\}$

Theorem: Assuming one-way functions exist, for any $n(\lambda) = \mathbf{poly}(\lambda)$, there exist one-time signatures with message length $n(\lambda)$!

- Key length: $|\mathrm{vk}| = n(\lambda) \cdot \lambda > n(\lambda)$ (long keys)
- We will fix this later
- We don't need any number theory!

"Zero-Time Signatures"



$\mathrm{Gen}(1^\lambda) \to (\mathrm{sk}, \mathrm{vk})$



m

vk (broadcast)

$(m, \sigma = \mathrm{Sign}(\mathrm{sk}, m))$



$\mathrm{Verify}(\mathrm{vk}, m, \sigma) \in \{0,1\}$

Definition: A "zero-time" digital signature scheme with message length $n = n(\lambda)$ consists of efficient algorithms $\mathbf{Gen}, \mathbf{Sign}, \mathbf{Verify}$. It must satisfy **correctness** as well as:

- "Zero-time" Existential Unforgeability: for all efficient adversaries $A(\mathrm{vk})$, we have:

$$\Pr_{(\mathrm{sk},\mathrm{vk})}[A(\mathrm{vk}) \to (m^*, \sigma^*): \mathrm{Verify}(\mathrm{vk}, m^*, \sigma^*) = 1] = \mathrm{negl}(\lambda)$$

“Zero-Time Signatures”



$\text{Gen}(1^\lambda) \rightarrow (x, f(x))$



m

$y = f(x)$

$(m, \sigma = x)$



$\text{Verify}(y, m, \sigma) = 1$ iff $f(\sigma) = y$

Construction from any OWF:

- $\text{Gen}(1^\lambda) \rightarrow (x, y = f(x))$
- To sign a message, send x. To verify, check that σ is a valid preimage of y.
- Unforgeability follows immediately from one-wayness of f

One-Time Signatures





$m \in \{0,1\}^n$



Each m now needs to have its own distinct signature.

One-Time Signatures

$\mathrm{Gen}(1^\lambda) \to \left((x_{i,b}), \left(f(x_{i,b})\right)\right)$





$(y_{i,b})_{i,b}$

$\left(m, \sigma = (x_{i,m_i})_i\right)$



$m \in \{0,1\}^n$

$\mathrm{Verify}(\mathrm{vk}, m, \sigma) = 1$ iff $f(\sigma_i) = y_{i,m_i}$

Construction from any OWF:

- $\mathrm{Gen}(1^\lambda) \to \left((x_{i,b})_{i\in[n],b\in\{0,1\}}, \left(f(x_{i,b})\right)_{i\in[n],b\in\{0,1\}}\right)$ ($2n$ input/output pairs)
- To sign a message, send n preimages: x_{i,m_i} for all i.
- To verify, check the n preimages are all valid.

One-Time Signatures

Security Proof: suppose that



$\mathrm{vk} = (y_{i,b})_{i,b}$

m

$(x_{i,m_i})_i$

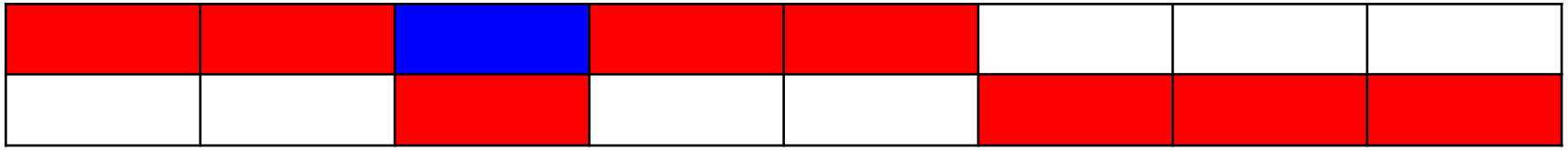
m^*, σ^*



$$\Pr_{(\mathrm{sk},\mathrm{vk})}\left[A^{\mathrm{Sign}(\mathrm{sk},\cdot)}(\mathrm{vk}) \to \text{valid } m^* \neq m, \sigma^*\right] = \epsilon$$

e.g.: $m = (0,0,1,0,0,1,1,1)$

$m_3^* = 0$



One-Time Signatures

Security Proof:

$y = f(x)$



sample i^*, b^*



$\mathrm{vk} = (y_{i,b})_{i,b}$

m

If $m_{i^*} \neq b^*$: $(x_{i,m_i})_i$

m^*, σ^*

$y_{i^*,b^*} = y$



Get m, m^*, σ^*

$\sigma^*_{i^*}$

$$\Pr[f(\sigma^*_{i^*}) = y] \geq \frac{\epsilon}{2n}$$

- Emulation succeeds with probability $1/2$
- Conditioned on forgery, $m^*_{i^*} = b^*$ with probability $\geq \frac{1}{n}$ (since $m^* \neq m$)

One-time signatures with short keys

One-Time Signatures



$\mathrm{Gen}(1^\lambda) \rightarrow (\mathrm{sk}, \mathrm{vk})$



vk (broadcast)

$(m, \sigma = \mathrm{Sign}(\mathrm{sk}, m))$

$m \in \{0,1\}^n$



$\mathrm{Verify}(\mathrm{vk}, m, \sigma) \in \{0,1\}$

- Suppose we start with a scheme with large vk
- Do PRGs help?
  - Can compress sk but not vk
- Not clear how to compress vk.

Hash-and-Sign Paradigm



$\text{Gen}(1^\lambda) \rightarrow (\text{sk}, \text{vk})$



$m \in \{0,1\}^n$

vk (broadcast)

$(m, \sigma = \text{Sign}(\text{sk}, m))$



$\text{Verify}(\text{vk}, m, \sigma) \in \{0, 1\}$

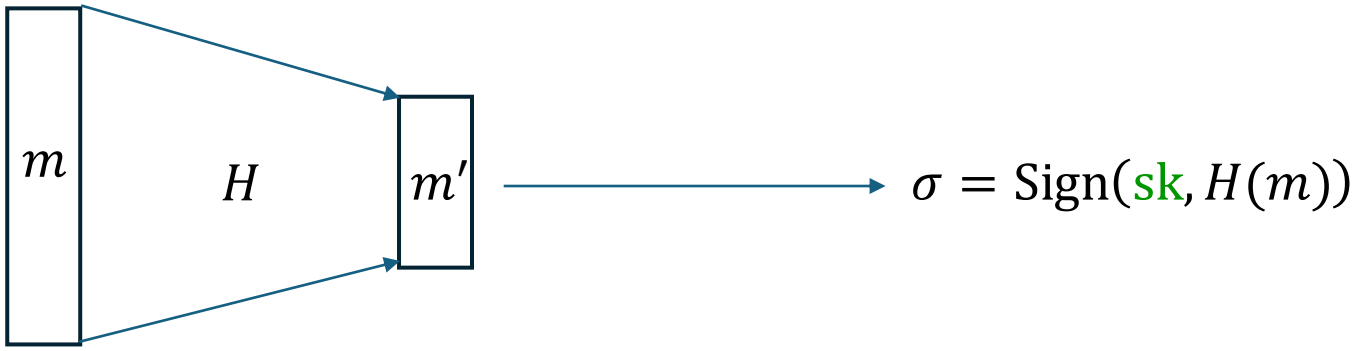
- Idea: extend the scheme to handle longer messages with (roughly) same vk



- H must be a publicly computable "hash function" (so that σ can be verified)

so (s

Hash-and-Sign Paradigm



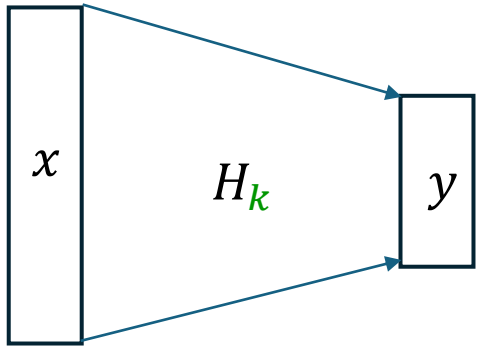
- Is this secure?
- If H is a fixed function: no!
  - Suppose that $H(m_1) = H(m_2)$. Then can forge on m_2 given signature on m_1.

Hash-and-Sign Paradigm



- Is this secure?
- If H is a fixed function: no!
  - Suppose that $H(m_1) = H(m_2)$. Then can forge on m_2 given signature on m_1.
- Instead: use a keyed hash function H_k
  - Add k to both the signing and verification keys

Collision-Resistant Hash Functions



Definition: A family of efficiently computable functions $\mathcal{H} = \{H_k : \{0,1\}^{n(\lambda)} \to \{0,1\}^{\lambda}\}_{k \in K}$ is a collision-resistant hash function (CRHF) family with input length $n = n(\lambda)$ if for all polynomial-time A,

$$\Pr[A(k) \to (x_1, x_2) : x_1 \neq x_2 \text{ and } H_k(x_1) = H_k(x_2)] = \text{negl}(\lambda)$$

Collision-Resistant Hash Functions



Theorem: If

- $(\text{Gen}, \text{Sign}, \text{Verify})$ is a (one time OR full) digital signature scheme with message length $n = n(\lambda)$, and
- $\mathcal{H}$ is a collision-resistant hash function (CRHF) family with input length n,

then $(\text{Sign} \circ \mathcal{H}, \text{Verify} \circ \mathcal{H})$ is also a (one time OR full) digital signature scheme with

$$\text{Gen}'(1^\lambda) \to ((\text{sk}, k), (\text{vk}, k))$$

Collision-Resistant Hash Functions



So if we can write down a CHRF family with short hash keys, we can sign (single) long messages with short $(\mathrm{sk}, \mathrm{vk})$!

Intuition: $H_k = \mathrm{PRF}_k$ would work… if k were secret.

Two kinds of constructions

- Algebraic CRHFs based on, e.g., discrete log
- "Obscure programs" approach with a heuristic analysis: random oracle model

CRHFs based on discrete log

$H_k : \mathbb{Z}_q \times \mathbb{Z}_q \to \mathbb{Z}_p$

$p = 2q + 1$

$k = (g, h)$

g generator for $\mathbf{QR}_p \subset \mathbb{Z}_p^\times$

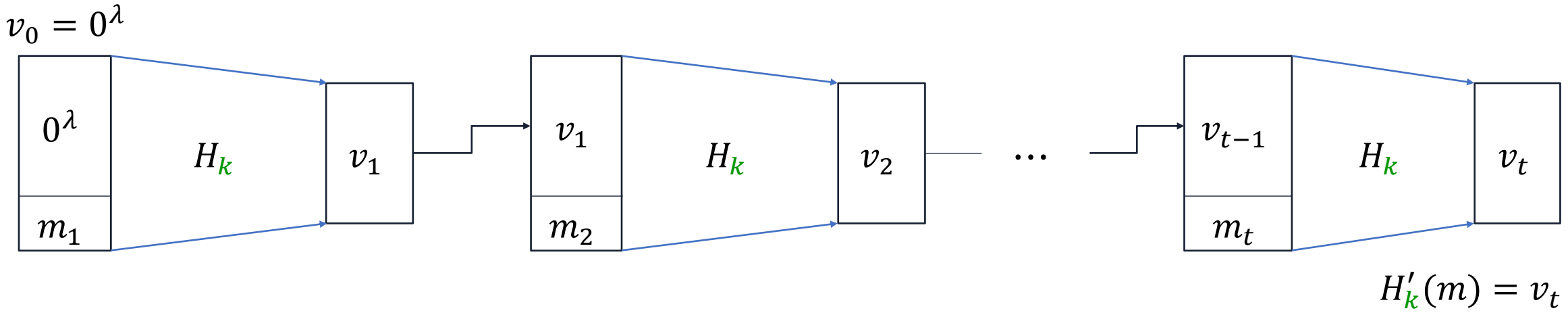
$h \leftarrow \mathbf{QR}_p$

$$H_k(x, y) = g^x h^y$$

CRHF Length Extension: Merkle–Damgård

$$H_k : \{0,1\}^{\lambda+1} \rightarrow \{0,1\}^{\lambda}$$

$$m = m_1 \parallel m_2 \parallel \cdots \parallel m_t, m_i \in \{0,1\}$$

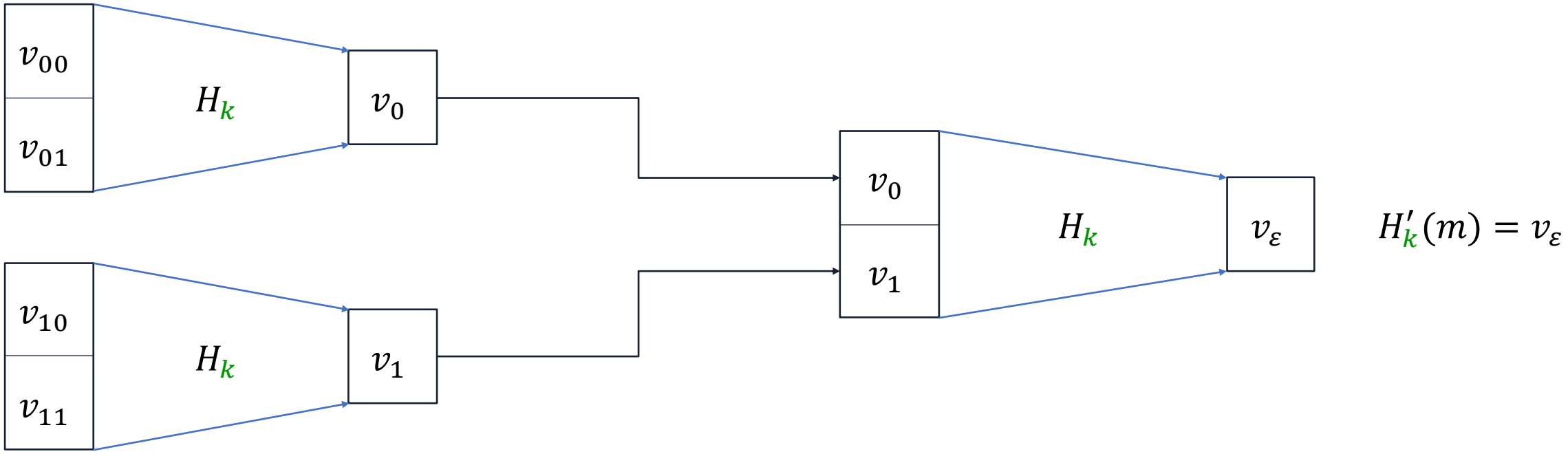


$$v_i = H_k(v_{i-1} \parallel m_i)$$

For fixed-length messages, each round processes one message bit.

CRHF Length Extension: Merkle Trees

$$H_k : \{0,1\}^{2\lambda} \rightarrow \{0,1\}^{\lambda}$$



$$m = v_{00} \parallel v_{01} \parallel v_{10} \parallel v_{11}, v_s \in \{0,1\}^{\lambda}$$

Four-block example with a fixed tree shape